plik keyboard.h

```
#ifndef KEYBOARD1_H
#define KEYBOARD1_H



char* DDR;
char* PORT;
char* PIN;
char* DDRLED;
char* PORTLED;

void led_init(char led_port);
void led_set(char output);
void set_IO_registers(char port);
void kb_init(char kb_port);
unsigned char get_key(char kb_size);

#endif
```

plik keyboard.c

```
#include "keyboard.h"
// #include <stdio.h>
#include <avr/io.h>
#include <util/delay.h>

//przypisanie zmiennym uniwersalnym odpowiedniego portu
void led_init(char led_port)

{
   if(led_port != 'a' || led_port != 'A' ||
   led_port != 'b' || led_port != 'B' ||
   led_port != 'c' || led_port != 'C' ||
   led_port != 'd' || led_port != 'D'){
       led_port = 'a';
   }
   set_IO_registers(led_port);
   DDRLED = DDR;
   PORTLED = PORT;
   *DDRLED = 0xFF; //ustawienie led�w na wyj�cie
}
```

```
void led_set(char output)
{
   *PORTLED = output;
}

//przypisanie zmiennym uniwersalnym odpowiedniego portu
void set_IO_registers(char port)
{
   switch (port)
   {
       case 'a':
       case 'A':
       DDR = &DDRA;
       PORT = &PORTA;
       PIN = &PINA;
       break;
       case 'b':
       case 'B':
       DDR = &DDRB;
       PORT = &PORTB;
       PIN = &PINB;

       break;
       case 'c':
       case 'C':
       DDR = &DDRC;
       PORT = &PORTC;
       PIN = &PINC;
       break;
       case 'd':
       case 'D':
       DDR = &DDRD;
       PORT = &PORTD;
       PIN = &PIND;
       break;
       default:
       DDR = &DDRB;
       PORT = &PORTB;
       PIN = &PINB;

       break;
   }
}
```

```
void kb_init(char kb_port)
{
   set_IO_registers(kb_port);
   *DDR = 0xF0;
   *PORT = 0x0F;
}

unsigned char get_key(char kb_size)
{
   char out = 0x00;
   //sprawdzenie wierszy
   *DDR = 0x0f;
   *PORT = 0xF0;
   char cols = *PIN;
   //je♦eli wiersz jest wci♦ni♦ty

   // sprawdzenie kolumn
   *DDR = 0xf0;
   *PORT = 0x0f;
   char rows = *PIN;
   int row =-1;
   int col=-1;
   int i;


       for( i = 0; i<4; i++){
           if(!(rows & (1 << i))){
               for(int j = 4; j<8; j++){
                   if(!(cols & (1 << (j)))){
                       row = i;
                       col = j-4;
                       break;
                   }
               }
               break;
           }
       }

   if(kb_size == 'm'){
       if (row != 0){
           return 0x00;
       }
```

```
    }

    if(row != -1 && col !=-1)
    {

        //out |= _BV(row);
        //out |= _BV(col);


        out = 4*col+row;

    }
    // }
    return out;
}
```

plik main.c

```
#include <avr/io.h>
#include <util/delay.h>
#include "keyboard.h"
#define F_CPU 16000000UL
int main (void){

    led_init('a');
    kb_init('b');

    while(1){
        led_set(get_key('w'));
        _delay_ms(50);
    }


    return 0;
}
```